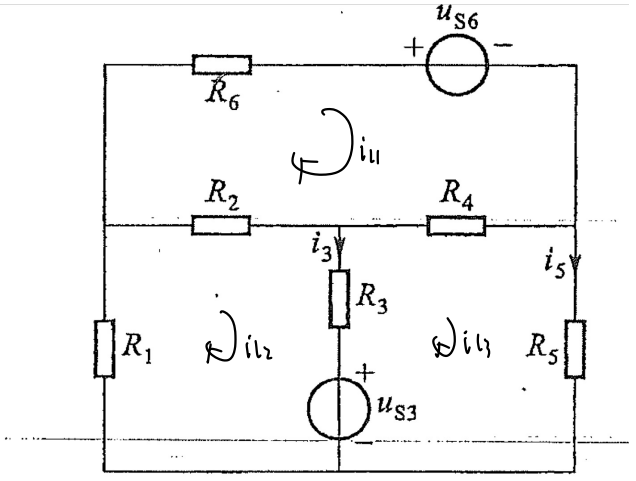
3-8 用网孔电流法求解题3-7图中电流 i_5。



题 3-7 图

$(R_2+R_4+R_6)i_{l1}-R_2i_{l2}-R_4i_{l3}=-u_{S6}$

$(R_2+R_3+R_1)i_{l2}-R_2i_{l1}-R_3i_{l3}=-u_{S3}$

$(R_4+R_5+R_3)i_{l3}-R_3i_{l2}-R_4i_{l1}=u_{S3}$

$\because R_1=R_2=10\Omega,\ R_3=4\Omega,\ R_4=R_5=8\Omega$

$R_6=2\Omega,\ u_{S3}=20V,\ u_{S6}=40V$

$\Rightarrow i_{l3}\approx -0.956A$

$\therefore i_5=-0.956A$

3-12 用回路电流法求解题3-12图所示电路中电流 I_α 及电压 U_o。

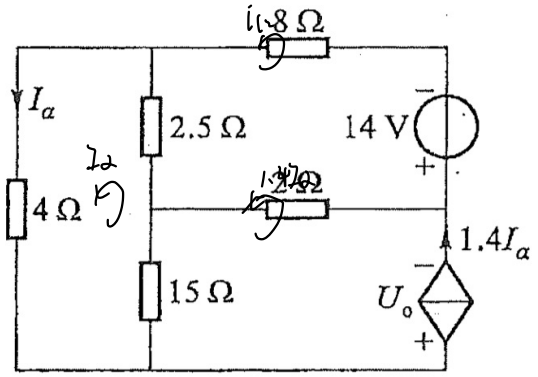
$i_{l1}=I_\alpha \quad i_{l3}=1.4I_\alpha$

$(4+2.5+15)I_\alpha-2.5i_{l2}-15\times1.4I_\alpha=0$

$(8+2.5+2)i_{l2}-2.5I_\alpha-2\times1.4I_\alpha=-14$

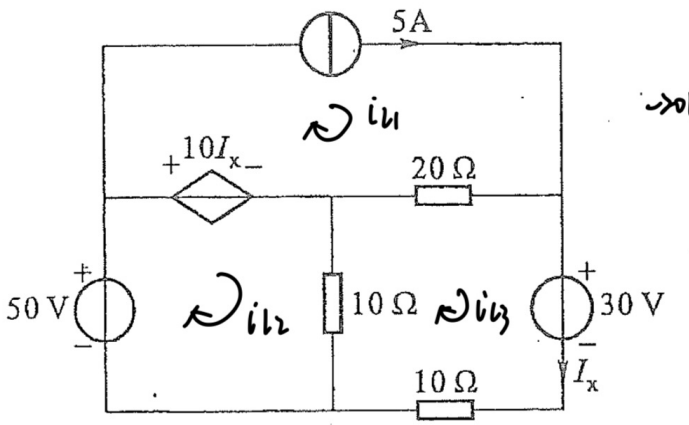
$U_o+(1.4I_\alpha-i_{l2})\times2+(1.4I_\alpha-I_\alpha)\times15=0$

$\Rightarrow I_\alpha=5A \qquad U_o=-42V$



题 3-12 图

3-14 用回路电流法求解题3-14图所示电路中 I_x 以及 CCVS 的功率。



$i_{l1}=5A \quad i_{l3}=I_x$

$10\times i_{l2}-10\times i_{l3}=50-10I_x$

$\Rightarrow 10i_{l1}-10\times i_{l2}+(10+20+10)i_{l3}=-30$

$\Rightarrow i_{l2}=5A,\ I_x=3A$

$I=i_{l2}-i_{l1}=0$

$\therefore P_{CCVS}=0$